Generative models

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Generative classifiers

Gaussian Discriminant Analysis

Naive Bayes

Discriminative vs Generative classifiers

Discriminative classifier

A discriminative classifier directly models the posterior $p(y|\boldsymbol{x})$. It can only be used to discriminate between classes.

Generative classifier

In contrast, a generative classifier models the class conditional density $p(\boldsymbol{x}|y)$. It can be used to generate examples $\boldsymbol{x}$ from each class y.
We can obtain the posterior using Bayes rule

$$p(y|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|y)p(y)}{p(\boldsymbol{x})}$$

We do not even need to calculate the denominator to make predictions:

$$\hat{y} = \arg\max_y p(\boldsymbol{x}|y)p(y)$$

Generative classifiers

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Model

In Gaussian discriminant analysis, the class confitional densities are multivariate Gaussians:

$$p(\boldsymbol{x}|y = c; \boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$$

The posterior has the following form:

$$p(y = c|\boldsymbol{x}; \boldsymbol{\theta}) \propto \pi_c \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$$

where $\pi_c = p(y = c)$ is the prior probability of label c

Quadratic decison boundaries

The log posterior over class label is

$$\log p(y = c|\boldsymbol{x}; \boldsymbol{\theta}) = \log \pi_c - \frac{1}{2} \log |\boldsymbol{\Sigma}_c| - \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_c) + cst$$

This is the discriminant function which is a quadratic function of $\boldsymbol{x}$. The model is called **quadratic discriminant analysis** or **QDA**.

Linear decison boundaries

If the covariance matrices are shared across classes, $\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$, the log posterior becomes

$$\begin{aligned}\log p(y = c|\boldsymbol{x}; \boldsymbol{\theta}) &= \log \pi_c - \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_c) + cst \\ &= \log \pi_c - \frac{1}{2}\boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c + \boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c - \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x} + cst \\ &= a_c + \boldsymbol{x}^\top b_c + cst\end{aligned}$$

The discriminant function is a linear function of $\boldsymbol{x}$. This is called **Linear discriminant analysis** or **LDA**.

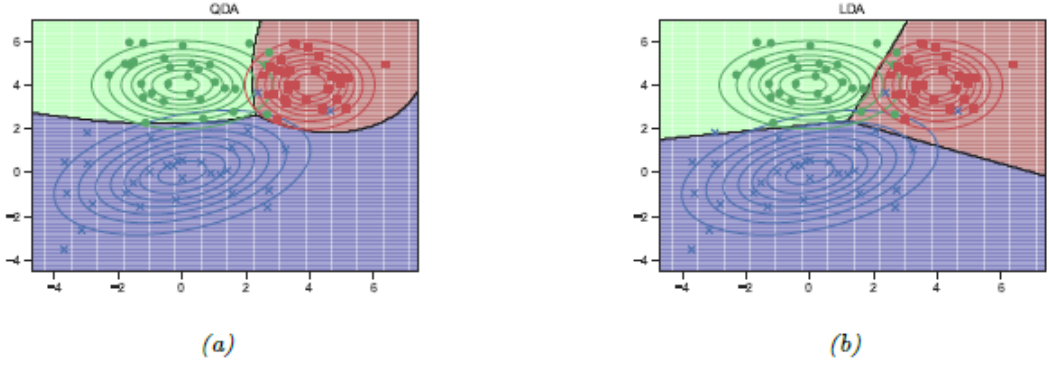


Figure: QDA and LDA fit to data from 3 classes

Fitting the model

Using MLE, the likelihood function is

$$\prod_{i=1}^{N} p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = \prod_{i=1}^{N} \text{Cat}(y_i|\boldsymbol{\pi}) \prod_{c=1}^{C} \mathcal{N}(\boldsymbol{x}_i;\boldsymbol{\mu}_c,\boldsymbol{\Sigma}_c)^{\mathbb{I}(y_i=c)}$$
$$= \prod_{i=1}^{N}\prod_{c=1}^{C} \pi_c^{\mathbb{I}(y_i=c)} \prod_{c=1}^{C} \mathcal{N}(\boldsymbol{x}_i;\boldsymbol{\mu}_c,\boldsymbol{\Sigma}_c)^{\mathbb{I}(y_i=c)}$$

the log-likelihood is

$$\left[\sum_{i=1}^{N}\sum_{c=1}^{C} \mathbb{I}(y_i=c)\log\pi_c\right] + \sum_{c=1}^{C}\left[\sum_{i:y_i=c} \log\mathcal{N}(\boldsymbol{x}_i;\boldsymbol{\mu}_c,\boldsymbol{\Sigma}_c)\right]$$

The parameters $\boldsymbol{\pi}$ and $(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ can be optimized separately. The MLE for the prior is

$$\hat{\pi}_c = \frac{N_c}{N}$$

The MLE for the Gaussians are as follows :

$$\hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{i:y_i=c} \boldsymbol{x}_i$$

$$\hat{\boldsymbol{\Sigma}}_c = \frac{1}{N_c} \sum_{i:y_i=c} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)^\top$$

where N_c is the number of observations whose class is c.
For LDA, $\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$, the covariance matrix estimate is

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{c=1}^{C} \sum_{i:y_i=c} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)^\top$$

Regularized discriminant analysis

MLE $\hat{\Sigma}_c$ can overfit if N_c is small compared to D. Forving the $\hat{\Sigma}_c$ to be diagonal can overcome the problem.
We can also use a MAP estimate of a shared full covariance matrix with an inverse Wishart prior (distribution over positive definite matrices). The MAP estimate is

$$\hat{\Sigma}_{MAP} = \lambda diag(\hat{\Sigma}_{MLE}) + (1 - \lambda)\hat{\Sigma}_{MLE}$$

where λ controls the regularization. This is called **Regularized discriminant analysis**. There are robusts ways to invert $\hat{\Sigma}_{MAP}$.

There is more to Gaussian discriminant analysis.

Explore further

- Diagonal covariances (Naive Bayes assumption)
- Shared diagonal covariance matrix (Diagonal LDA)
- Nearest centroid classifier (nearest class mean classifier)
- Nearest class mean metric learning
- Fisher's linear discriminant analysis
- Connection between LDA and logistic regression

Generative classifiers

Gaussian Discriminant Analysis

Naive Bayes

Model

It is a generative classifier which assumes that the features are conditionally independent given the class label (Naive Bayes assumption). The model is called "naive" since the features are generally not independent still it gives good results. The class conditional density has the form

$$p(\boldsymbol{x}|y=c;\boldsymbol{\theta}) = \prod_{d=1}^{D} p(x_d|y=c;\boldsymbol{\theta}_{dc})$$

where $\boldsymbol{\theta}_{dc}$ are the parameters for class c and feature d. The posterior is given by

$$p(y=c|\boldsymbol{x};\boldsymbol{\theta}) = \frac{\pi_c \prod_{d=1}^{D} p(x_d|y=c;\boldsymbol{\theta}_{dc})}{\sum_{c'} \pi'_c \prod_{d=1}^{D} p(x_d|y=c';\boldsymbol{\theta}_{dc'})}$$

where $\pi_c = p(y=c)$ is the prior probability of class c

Model

The distribution depends on the nature of x_d

Bernoulli

Bernoulli distribution for Binary features, $x_d \in \{0, 1\}$:
$p(\boldsymbol{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^{D} \text{Ber}(x_d; \theta_{dc})$ where θ_{dc} is the probability that $x_d = 1$ in class c.

Categorical

Categorical distribution for categorical features, $x_d \in \{1, ..., K\}$:
$p(\boldsymbol{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^{D} \text{Cat}(x_d; \boldsymbol{\theta}_{dc})$ where θ_{dck} is the probability that $x_d = k$ in class c.

Univariate Gaussian

Univariate Gaussian distribution for real-valued features, $x_d \in \{1, ..., K\}$:
$p(\boldsymbol{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^{D} \mathcal{N}(x_d; \mu_{dc}, \sigma_{dc}^2)$ where μ_{dck} and σ_{dc}^2 are the mean and variance of feature d when the class label is c (equivalent to Gaussian discriminant analysis using diagonal covariance matrices).

Model fitting

The MLE can be written as follows :

$$\prod_{i=1}^{N} p(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = \prod_{i=1}^{N} \text{Cat}(y_i|\boldsymbol{\pi}) \prod_{d=1}^{D} p(x_{id}|y_i;\boldsymbol{\theta}_d)$$
$$= \prod_{i=1}^{N} \text{Cat}(y_i|\boldsymbol{\pi}) \prod_{d=1}^{D} \prod_{c=1}^{C} p(x_{id};\boldsymbol{\theta}_{dc})^{\mathbb{I}(y_i=c)}$$

The log-likelihood is

$$\left[\sum_{i=1}^{N}\sum_{c=1}^{C} \mathbb{I}(y_i = c) \log \pi_c\right] + \sum_{c=1}^{C}\sum_{d=1}^{D}\left[\sum_{i:y_i=c} \log p(x_{id};\boldsymbol{\theta}_{dc})\right]$$

We can separately estimate the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\theta}_{dc}$. The MLE of $\boldsymbol{\pi}$ is

$$\hat{\pi}_c = \frac{N_c}{N}$$

The MLE of $\boldsymbol{\theta}_{dc}$ depends on the distribution of the class conditional density :

- Bernoulli :

$$\hat{\theta}_{dc} = \frac{N_{dc}}{N_c}$$

- Categorical :

$$\hat{\theta}_{dck} = \frac{N_{dck}}{\sum_{k'=1}^{K} N_{dck'}} = \frac{N_{dck}}{N_c}$$

  where N_{dck} is the number of times feature d had value k in examples of class c

- Univariate Gaussian :

$$\hat{\boldsymbol{\mu}}_{dc} = \frac{1}{N_c} \sum_{i:y_i=c} x_{id}$$

$$\hat{\sigma}^2_{dc} = \frac{1}{N_c} \sum_{i:y_i=c} (x_{id} - \hat{\mu}_{dc})^2$$

More

As usual, there is always more.

Explore further

- Bayesian naive Bayes
- Connection between naive Bayes and logistic regression